

Testrix — SRS AI Test Case Database Schema

Recommended MySQL schema for the workflow: SRS upload → Gemini test-case generation → Quick Review/Edit → approval → execution. The existing execution tables remain separate.

1. srs_documents

```
CREATE TABLE srs_documents (
  id BIGINT PRIMARY KEY AUTO_INCREMENT,
  file_name VARCHAR(255) NOT NULL,
  file_type VARCHAR(50) NOT NULL,
  file_size BIGINT NULL,
  file_path TEXT NULL,
  status ENUM('UPLOADED','PROCESSING','COMPLETED','FAILED')
    NOT NULL DEFAULT 'UPLOADED',
  total_chunks INT DEFAULT 0,
  total_requirements INT DEFAULT 0,
  total_test_cases INT DEFAULT 0,
  error_message TEXT NULL,
  uploaded_by BIGINT NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    ON UPDATE CURRENT_TIMESTAMP,
  INDEX idx_srs_uploaded_by (uploaded_by),
  INDEX idx_srs_status (status),
  INDEX idx_srs_created_at (created_at)
);
```

2. srs_requirements

```
CREATE TABLE srs_requirements (
  id BIGINT PRIMARY KEY AUTO_INCREMENT,
  srs_document_id BIGINT NOT NULL,
  requirement_code VARCHAR(100) NULL,
  title VARCHAR(500) NOT NULL,
  description TEXT NOT NULL,
  requirement_type ENUM(
    'FUNCTIONAL','NON_FUNCTIONAL','BUSINESS_RULE',
    'CONSTRAINT','ASSUMPTION','DEPENDENCY'
  ) DEFAULT 'FUNCTIONAL',
  actor VARCHAR(255) NULL,
  module VARCHAR(255) NULL,
  priority ENUM('HIGH','MEDIUM','LOW') NULL,
  status ENUM('LIVE','PARTIAL','PLANNED') NULL,
  source_section VARCHAR(500) NULL,
  source_text TEXT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (srs_document_id) REFERENCES srs_documents(id)
    ON DELETE CASCADE,
  INDEX idx_requirement_document (srs_document_id),
  INDEX idx_requirement_code (requirement_code)
);
```

3. test_cases

```
CREATE TABLE test_cases (
  id BIGINT PRIMARY KEY AUTO_INCREMENT,
  srs_document_id BIGINT NULL,
  requirement_id BIGINT NULL,
  test_case_code VARCHAR(100) NOT NULL,
  title VARCHAR(500) NOT NULL,
  description TEXT NULL,
  test_type ENUM(
    'FUNCTIONAL','NEGATIVE','BOUNDARY','VALIDATION',
    'BUSINESS_RULE','WORKFLOW','INTEGRATION','SECURITY',
    'NOTIFICATION','ERROR_HANDLING','USABILITY','COMPATIBILITY'
  ) NOT NULL DEFAULT 'FUNCTIONAL',
  priority ENUM('HIGH','MEDIUM','LOW') NULL,
  review_status ENUM(
    'AI_GENERATED','UNDER_REVIEW','APPROVED','REJECTED'
  ) NOT NULL DEFAULT 'AI_GENERATED',
  source_section VARCHAR(500) NULL,
  source_text TEXT NULL,
  created_by BIGINT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (srs_document_id) REFERENCES srs_documents(id)
```

```

        ON DELETE SET NULL,
        FOREIGN KEY (requirement_id) REFERENCES srs_requirements(id)
        ON DELETE SET NULL,
        INDEX idx_testcase_document (srs_document_id),
        INDEX idx_testcase_requirement (requirement_id),
        INDEX idx_testcase_review_status (review_status),
        INDEX idx_testcase_priority (priority),
        UNIQUE KEY uq_test_case_code (test_case_code)
    );

```

4. test_case_steps

```

CREATE TABLE test_case_steps (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    test_case_id BIGINT NOT NULL,
    step_number INT NOT NULL,
    action TEXT NOT NULL,
    expected_result TEXT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
        ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (test_case_id) REFERENCES test_cases(id)
        ON DELETE CASCADE,
    UNIQUE KEY uq_test_case_step (test_case_id, step_number),
    INDEX idx_steps_test_case (test_case_id)
);

```

5. test_case_preconditions

```

CREATE TABLE test_case_preconditions (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    test_case_id BIGINT NOT NULL,
    precondition TEXT NOT NULL,
    sequence_number INT DEFAULT 1,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (test_case_id) REFERENCES test_cases(id)
        ON DELETE CASCADE,
    INDEX idx_precondition_test_case (test_case_id)
);

```

6. test_case_test_data

```

CREATE TABLE test_case_test_data (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    test_case_id BIGINT NOT NULL,
    data_key VARCHAR(255) NULL,
    data_value TEXT NULL,
    description TEXT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
        ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (test_case_id) REFERENCES test_cases(id)
        ON DELETE CASCADE,
    INDEX idx_test_data_test_case (test_case_id)
);

```

7. ai_test_generation_runs

```

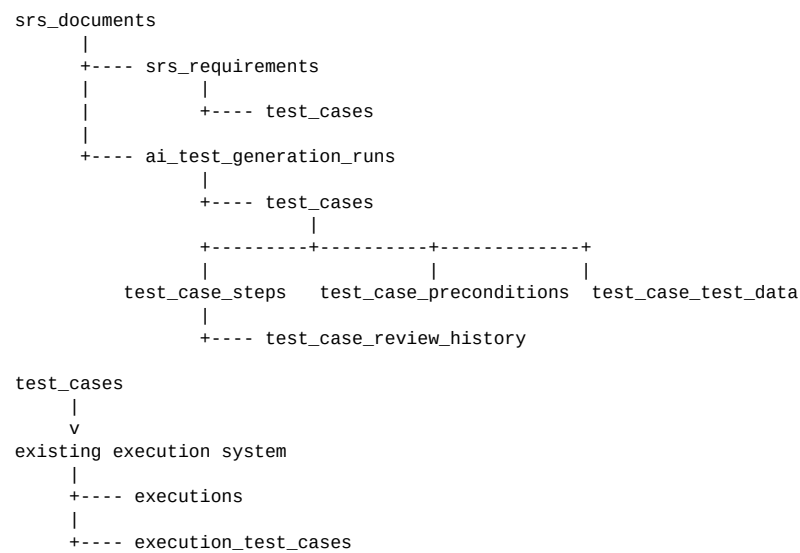
CREATE TABLE ai_test_generation_runs (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    srs_document_id BIGINT NOT NULL,
    provider VARCHAR(50) NOT NULL DEFAULT 'GEMINI',
    model VARCHAR(100) NULL,
    status ENUM('STARTED', 'PROCESSING', 'COMPLETED', 'FAILED')
        NOT NULL DEFAULT 'STARTED',
    total_chunks INT DEFAULT 0,
    processed_chunks INT DEFAULT 0,
    generated_test_cases INT DEFAULT 0,
    duplicate_test_cases INT DEFAULT 0,
    final_test_cases INT DEFAULT 0,
    processing_time_ms BIGINT NULL,
    error_message TEXT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    completed_at TIMESTAMP NULL,
    FOREIGN KEY (srs_document_id) REFERENCES srs_documents(id)
        ON DELETE CASCADE,
    INDEX idx_ai_run_document (srs_document_id),
    INDEX idx_ai_run_status (status)
);

```

8. test_case_review_history

```
CREATE TABLE test_case_review_history (
  id BIGINT PRIMARY KEY AUTO_INCREMENT,
  test_case_id BIGINT NOT NULL,
  action ENUM(
    'CREATED', 'EDITED', 'APPROVED', 'REJECTED', 'RESTORED'
  ) NOT NULL,
  field_name VARCHAR(100) NULL,
  old_value TEXT NULL,
  new_value TEXT NULL,
  changed_by BIGINT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (test_case_id) REFERENCES test_cases(id)
    ON DELETE CASCADE,
  INDEX idx_review_history_test_case (test_case_id),
  INDEX idx_review_history_created_at (created_at)
);
```

Architecture & Table Responsibilities



Important Design Rule

test_cases stores the test definition generated/edited by QA. **execution_test_cases** stores what happened when that test was executed. Do not replace the existing execution table with the new test_cases table.

Quick Review / Excel-like Editing

Use test_cases.review_status for the review workflow: AI_GENERATED → UNDER_REVIEW → APPROVED or REJECTED. Use test_case_review_history to retain edits and approval/rejection history. test_case_steps keeps individual steps editable instead of storing them as one large text field.